

Go-Around Pipeline — Terminal Cheat Sheet

Copy-paste commands for the Mac Terminal. Every command assumes you are inside the project folder (step 0).

0. Always start here (every new Terminal window)

```
cd ~/Documents/GA_algorithm
```

1. Run the analysis on a dataset

Basic run (dataset = folder name inside datasets/):

```
.venv/bin/python run_analysis.py KDAB_2025
```

If there is only ONE dataset folder, you can skip the name:

```
.venv/bin/python run_analysis.py
```

Useful variants:

```
.venv/bin/python run_analysis.py KDAB_2025 --report      # + figures, report, PDF
.venv/bin/python run_analysis.py KDAB_2025 --calibrate   # + cutoff-check histogram
.venv/bin/python run_analysis.py KDAB_2025 --no-plots    # faster, skips trajectory plots
.venv/bin/python run_analysis.py KDAB_2025 --limit 5     # quick test: first 5 days only
```

Results appear in a NEW folder each time: output/<dataset>/run_01/, run_02/, ... (events CSV, all approaches CSV, summary.pdf + summary tables, plots/, run_config.txt).

2. Add a new dataset (existing airport)

Make a folder in datasets/ named ICAO_anything, drop the daily .parquet/.csv files in, run it:

```
mkdir -p datasets/KDAB_2026
# (copy your daily files into datasets/KDAB_2026 using Finder or:)
cp /path/to/your/files/*.parquet datasets/KDAB_2026/
.venv/bin/python run_analysis.py KDAB_2026
```

3. Set up a NEW airport (e.g. Nashville, KBNA)

Step 1 — generate the airport profile (internet needed once):

```
.venv/bin/python run_analysis.py new-airport KBNA
```

Step 2 — open the generated profile and review it (set timezone, preset, terrain, arrivals flag):

```
open -e airports/KBNA.yaml
```

Inside the file set: timezone (e.g. America/Chicago) - preset (training_ga or air_carrier) - terrain (flat or dem) - assume_arrivals_dataset (true only for arrivals-query data). Save and close.

Step 3 — create the dataset folder, add data, first run WITH the calibration check:

```
mkdir -p datasets/KBNA_2026
# (copy daily files into datasets/KBNA_2026, then:)
.venv/bin/python run_analysis.py KBNA_2026 --calibrate
```

Then open calibration_plateau.png in the run folder: the shaded band must sit in an empty gap between the colored populations. Spot-check a few trajectory plots in plots/ before trusting the numbers.

4. One-time setup on a new Mac (only if .venv is missing)

```
cd ~/Documents/GA_algorithm
```

```
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt
```

5. Handy extras

```
open output                                # browse results in Finder
open output/KDAB_2025/run_01/summary.pdf   # read the summary
.venv/bin/python -m goaround_pipeline.sensitivity # threshold sweep
.venv/bin/python -m goaround_pipeline.validate_traffic 20250108 # cross-check
```